

STANDARD OPERATING PROCEDURE

SOP NO.: MHPL/VL/004-02	RECOVERY STUDY	Date Prepared.: - 17/09/2025
		Effective Date.: - 10/10/2025
Next review Date.: - 09/10/2027		
Supersedes.: MHPL/VL/004-01		Page.: 1 of 4

1.0 PURPOSE

To define the procedure for conducting a Recovery Study in areas serviced by HVAC systems, including cleanrooms, LAF units, and dynamic pass boxes, to evaluate how quickly these areas return to specified cleanliness levels after particle loading.

2.0 SCOPE

This study applies to non-unidirectional airflow cleanrooms only. It evaluates how quickly the room recovers to clean conditions after particle contamination, based on air dilution and mixing. It is not suitable for unidirectional airflow systems, where particles are removed by one-way airflow. Key factors include air distribution, temperature, and obstructions.

3.0 RESPONSIBILITIES

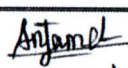
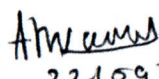
Validation Engineer/Service Engineer/Service Technician.

4.0 DEFINITIONS**Recovery Time:**

The time required for a controlled environment to return to its specified cleanliness class after a contamination event.

5.0 PROCEDURE**5.1 Recovery Test for Classified Area**

- 5.1.1** Ensure the AHU (Air Handling Unit) of the classified area at least 30 minutes before beginning the test to allow stabilization.
- 5.1.2** Record the at-rest airborne particle counts by measuring particles at designated sampling points throughout the room.
- 5.1.3** Select sampling locations that represent actual operational areas, such as typical working positions of personnel at approximately 900 mm (3 feet) above floor level. Also include points in

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front of the center of return air riser grilles, about 300 mm from the grille face, with the sampling probe facing the airflow, or after measuring particle counts, identify the location with the highest particle concentration and designate it as the recovery test location.

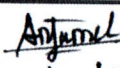
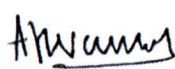
- 5.1.4 Set up the particle counter according to the manufacturer's instructions and ensure it has a valid calibration certificate.
- 5.1.5 Use a particle size channel less than 1 μm . (or as per customer requirement), It is recommended to select the channel corresponding to the maximum aerosol concentration detectable by the LSAPC (Light Scattering Airborne Particle Counter).
- 5.1.6 Take the initial particle count reading, then generate smoke and begin measuring particle counts at intervals of no more than 1 minute.
- 5.1.7 Contaminate the cleanroom or clean zone with an artificial aerosol while the air-handling units (AHUs) are in normal operation.
- 5.1.8 Increase the initial particle concentration to 10 or 100 times higher than the target cleanliness level, depending on the classification or specific requirements.
- 5.1.9 Particle counter give printout for both time and particle concentration at each 1 min interval.
- 5.1.10 Once the particle concentration reaches the limit specified by the classification (or as per customer protocol) record the time as the recovery time.
- 5.1.11 For the next location, repeat the procedure as previously described.

5.1.12 Acceptance Criteria:

The area must achieve its at-rest classified particle count limits (as per ISO or GMP standards) within 15 minutes of removing the smoke source.

5.2 Recovery Test for Equipment (if, Required):

- 5.2.1 Ensure the Equipment is ON condition and operating under stable conditions before initiating the recovery study.
- 5.2.2 Identify and select the location with the highest Non-Viable Particle Count (NVPC) to serve as the designated sampling point for the recovery test.

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5.2.3 Identify and select the location with the highest Non-Viable Particle Count (NVPC), or divide the working zone of the LAF unit or Dispensing Booth into a grid layout with sections measuring approximately 3 ft × 3 ft, to serve as the designated sampling point for the recovery test.

5.2.4 Set up the particle counter according to the manufacturer's instructions and ensure it has a valid calibration certificate.

5.2.5 Use a particle size channel less than 1 µm. (or as per customer requirement), It is recommended to select the channel corresponding to the maximum aerosol concentration detectable by the LSAPC (Light Scattering Airborne Particle Counter).

5.2.6 Take the initial particle count reading, then generate smoke and begin measuring particle counts at intervals of no more than 1 minute.

5.2.7 Contaminate the equipment with an artificial aerosol or switch off the equipment to create contamination while it is in normal operation.

5.2.8 Increase the initial particle concentration to 10 or 100 times higher than the target cleanliness level, depending on the classification or specific requirements.

5.2.9 Particle counter give printout for both time and particle concentration at each 1 min interval.

5.2.10 Once the particle concentration reaches the limit specified by the classification (or as per customer protocol) record the time as the recovery time.

5.2.11 For the next location, repeat the procedure as previously described.

5.2.12 Acceptance criteria:

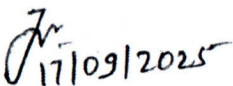
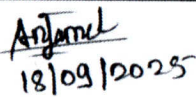
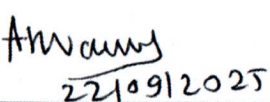
The area must achieve particle count limits compliant with ISO 5 (Grade A) classification within a maximum of 10 minutes.

5.3 FREQUENCY

Once in a Two year or as per customer requirement.

6.0 ANNEXURE

Annexure I: Recovery Test (F/VL/004/01-01)

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7.0 REFERENCE

ISO14644-3 2019 (E): Cleanrooms and associated controlled environments Part-3 Metrology and test methods.

8.0 ABBREVIATION

NVPC: Non-Viable Particle Count

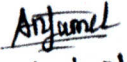
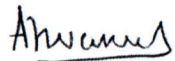
ISO: International Organization for Standardization

HEPA: High Efficiency Particulate Air

LSAPC: Light Scattering Airborne Particle Counter

9.0 REVISION HISTORY

Revision No.	Effective Date	Reasons for Revision
00	21/10/2021	First Issue
01	15/10/2023	Periodic review
02	10/10/2025	Periodic review & Procedure updated for suitable operation.

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RECOVERY TEST

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Annexure I

AHU No./Equipment Id No.:	Date:
Room Name:	Room No:
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